

◆ AWS CLOUD PRACTITIONER BOOTCAMP

THURSDAY — SESSION 2

Shared Responsibility + Risk Awareness

Student Learning Notes (Expanded Edition)

◆ SESSION OBJECTIVE

By the end of this session, students should understand:

- ✓ What the AWS Shared Responsibility Model is
 - ✓ The difference between AWS responsibilities and customer responsibilities
 - ✓ Why most cloud security incidents happen
 - ✓ How misconfiguration creates risk
 - ✓ Why governance matters from the beginning of a cloud journey
 - ✓ Basic real-world cloud security scenarios
 - ✓ How cloud responsibility changes across AWS services
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◆ 1. UNDERSTANDING CLOUD SECURITY

Cloud computing changes how infrastructure is managed, but it does **not remove responsibility**.

One of the biggest misconceptions in cloud computing is:

“AWS handles all security.”

This is false.

AWS secures the cloud platform itself, but customers are responsible for how they use that platform.

This concept is known as:

The Shared Responsibility Model

This is one of the most important foundational concepts in AWS.

◆ 2. WHY THE SHARED RESPONSIBILITY MODEL EXISTS

Traditional data centers required companies to:

- Buy physical servers
- Secure buildings
- Manage hardware
- Maintain networking devices
- Handle cooling and electricity

AWS removes much of this operational burden.

However:

Even though AWS manages infrastructure, customers still control:

- Their data
- Their users
- Their permissions
- Their applications
- Their configurations

This division of responsibility creates:

- Flexibility
 - Scalability
 - Shared accountability
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◆ 3. AWS RESPONSIBILITY

“Security OF the Cloud”

AWS is responsible for protecting the infrastructure that runs AWS services.

This includes:

Physical Security

AWS protects:

- Data centers
- Buildings
- Physical servers
- Hardware devices

AWS data centers are highly secure facilities with:

- Restricted access
 - Surveillance systems
 - Environmental controls
 - Redundant power systems
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Infrastructure Security

AWS manages:

- Server hardware
- Networking infrastructure
- Storage infrastructure
- Global backbone networks

AWS also maintains:

- Availability Zones
 - Regions
 - Core cloud platform services
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Managed Service Infrastructure

For managed services, AWS handles:

- Infrastructure maintenance
- Hardware replacement
- Core platform patching

Examples:

- RDS infrastructure
- Lambda runtime infrastructure
- DynamoDB infrastructure

◆ 4. CUSTOMER RESPONSIBILITY

“Security IN the Cloud”

Customers are responsible for everything they place inside AWS.

This includes:

Identity & Access Management (IAM)

Customers control:

- Users
- Roles
- Permissions
- Authentication methods

Important:

Weak IAM controls are one of the biggest cloud risks.

Examples of bad IAM practices:

- Shared accounts
- No MFA
- Excessive permissions
- Using root account daily

Data Protection

Customers are responsible for:

- Encryption decisions
- Sensitive data handling
- Backup strategies
- Data classification

AWS provides tools for encryption, but customers decide:

- Whether encryption is enabled
- Who can access data

- Which keys are used
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Operating Systems & Applications

For services like EC2:

Customers manage:

- Operating system patching
- Installed software
- Firewall rules
- Security updates

AWS does not patch customer applications.

Network Configuration

Customers configure:

- Security Groups
- Network ACLs
- VPC design
- Internet exposure

Misconfigured networking can expose systems directly to the internet.

◆ 5. THE MOST IMPORTANT PRINCIPLE

AWS secures the cloud.

Customers secure what they put inside the cloud.

This distinction is critical for:

- Real-world cloud operations
 - Security management
 - AWS certification exams
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◆ 6. RESPONSIBILITY CHANGES BY SERVICE

Not all AWS services work the same way.

As AWS manages more infrastructure, customer operational effort decreases.

However:

Reduced management does NOT mean reduced accountability.

◆ 7. EC2 — INFRASTRUCTURE AS A SERVICE (IaaS)

Amazon EC2 provides virtual servers.

With EC2, customers have significant control.

AWS Manages

- Physical servers
 - Storage hardware
 - Networking backbone
 - Virtualization layer
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Customer Manages

- Operating system
 - Applications
 - Security patches
 - Firewall rules
 - IAM access
 - Data protection
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Important Insight

EC2 gives maximum flexibility, but also maximum responsibility.

The customer controls almost everything above the virtualization layer.

◆ 8. RDS — MANAGED DATABASE SERVICE

Amazon RDS is a managed database service.

AWS automates much of the infrastructure management.

AWS Manages

- Database infrastructure
 - OS patching
 - Automated backups (if enabled)
 - Database availability
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Customer Manages

- Database users
 - Database permissions
 - Data encryption
 - Data classification
 - Network access
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Important Insight

Managed services reduce operational burden, but customers still own data security.

◆ 9. S3 — STORAGE SERVICE

Amazon S3 stores objects and files.

AWS secures the storage infrastructure itself.

Customers control:

- Bucket permissions
 - Public access settings
 - Access policies
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Real-World Risk

Many companies accidentally exposed sensitive data because:

- S3 buckets were left public
 - Policies were misconfigured
 - Access permissions were too broad
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Important Principle

AWS did not expose the data.

The customer configuration exposed the data.

◆ 10. LAMBDA — SERVERLESS COMPUTING

AWS Lambda allows code execution without managing servers.

AWS handles:

- Servers
- Runtime infrastructure
- Scaling

Customers still manage:

- IAM permissions
 - Application logic
 - API security
 - Sensitive data handling
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Important Insight

Serverless reduces infrastructure management — not security responsibility.

◆ 11. HOW CLOUD RISK ACTUALLY STARTS

Most cloud security incidents do not happen because AWS infrastructure failed.

Most happen because of:

- Human error
 - Weak permissions
 - Misconfiguration
 - Lack of governance
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◆ 12. WHAT IS MISCONFIGURATION?

Misconfiguration means:

Incorrectly configuring cloud resources in a way that creates risk.

Examples:

- Public S3 buckets
 - Open security groups
 - Excessive IAM permissions
 - Weak passwords
 - Missing MFA
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◆ 13. REAL-WORLD SCENARIO — PUBLIC S3 BUCKET

Situation

A company uploads sensitive documents to Amazon S3.

Failure

The S3 bucket is configured as public.

Result

Anyone on the internet can access the files.

Possible Impact

- Data leaks
 - Compliance violations
 - Financial penalties
 - Reputation damage
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Key Lesson

AWS secured the infrastructure.

The customer failed to secure access.

◆ 14. REAL-WORLD SCENARIO — OPEN SECURITY GROUP

Situation

An EC2 server is deployed.

Failure

SSH port (22) is open to:
0.0.0.0/0

This means:
Any IP address can attempt connection.

Result

Attackers attempt:

- Password guessing
- Brute-force attacks
- Unauthorized access

Key Lesson

Every open port expands the attack surface.

◆ 15. REAL-WORLD SCENARIO — WEAK IAM CONTROLS

Situation

A team shares one administrator account.

Failure

- No MFA
- Shared credentials
- No role separation

Result

If credentials are stolen:
The attacker gains full access.

Key Lesson

Identity is one of the most important security layers in cloud computing.

◆ 16. WHAT IS GOVERNANCE?

Governance means:

Establishing rules, standards, controls, and visibility across cloud environments.

Governance helps organizations:

- Maintain consistency
- Reduce risk
- Improve visibility
- Standardize deployments
- Enforce security policies

◆ 17. WHY GOVERNANCE MATTERS EARLY

Without governance:

Organizations often experience:

- Random configurations
- Security gaps
- Operational chaos
- Limited visibility

With Governance

Organizations gain:

- Standardized environments
- Controlled access
- Better monitoring
- Reduced human error
- Improved scalability

◆ 18. GOVERNANCE IS NOT “RESTRICTION”

Many beginners think governance slows innovation.

In reality:

Good governance enables:

- Safer scaling
- Predictable operations
- Faster recovery
- Better security

◆ 19. KEY SECURITY CONCEPTS FOR BEGINNERS

Principle of Least Privilege

Users should only receive permissions required for their tasks.

MFA (Multi-Factor Authentication)

Adds an additional security layer beyond passwords.

Strongly recommended for:

- Root accounts
 - Admin accounts
 - Privileged users
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Attack Surface

Any exposed system component that attackers can target.

Examples:

- Open ports
 - Public IP addresses
 - Weak credentials
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Visibility

Organizations cannot protect resources they cannot see or monitor.

◆ 20. AWS CLOUD PRACTITIONER EXAM FOCUS

For the exam, students should remember:

AWS Responsibility

(Security OF the Cloud)

- Hardware
 - Infrastructure
 - Physical facilities
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Customer Responsibility

(Security IN the Cloud)

- Data
 - IAM
 - Applications
 - Operating systems
 - Network settings
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◆ 21. FINAL TAKEAWAYS

- ✓ Cloud security is shared
- ✓ Responsibility changes depending on service type
- ✓ Misconfiguration is one of the largest cloud risks
- ✓ IAM is critical to cloud security
- ✓ Governance creates structure and reduces risk
- ✓ AWS provides tools — customers must configure them correctly

◆ FINAL PRINCIPLE

Cloud platforms provide capability.

Governance and security determine whether that capability becomes an advantage or a liability.